Rebuttal for CoRL Submission #18

We thank the reviewers and Area Chair for identifying four concrete issues. We agree that the original learned loss weight and whole-lowdim mask require correction, and we separate the operator-held evidence below from the still missing fixed-mount control.

TABLE I
LATEST OPERATOR-REPORTED, OPERATOR-HELD SWIPE RESULTS. SUCCESS IS ONE COMPLETE ROLLER ROTATION WITHIN 60 S. THESE PRELIMINARY COUNTS ARE NOT POOLED WITH FIXED-MOUNT TRIALS; WILSON 95% INTERVALS ARE SHOWN.

Policy or isolated change	Success/N	95% CI
Legacy Full	$9/10$	$[.596, .982]$
Phase-DP†	$12/20$	$[.387, .781]$
Progress-DP†	$14/20$	$[.481, .855]$
Learned weight $\rightarrow$ fixed $1+c$	$15/25$	$[.407, .766]$
Whole-lowdim $\rightarrow$ tactile-only mask	$13/20$	$[.433, .819]$
No-gating retrain	$8/20$	$[.219, .613]$
Unbounded $\rightarrow$ bounded gate	$7/20$	$[.181, .567]$

Q1: Contact anticipation versus task progress. Phase-DP and Progress-DP use the same diffusion backbone and add only a four-phase or normalized-progress condition. VPCE instead predicts a continuous 15-step future-contact horizon online from current wrist RGB and recent robot motion; unlike a hierarchical policy, it neither switches sub-policies nor uses a discrete high-level controller. Because the archived real-time fallback for Phase/Progress advanced its private clock faster than physical control time, the rows marked † remain preliminary pending an elapsed-time audit and an aligned N=20 rerun; we make no policy-level superiority claim from them.

As a predictor-level diagnostic on held-out trajectories, however, VPCE is not explained by progress alone: Progress captures some timing, but vision, motion, and tactile-derived contact supervision add information beyond normalized trajectory time. We nevertheless do not claim a universal transferable representation: source-teacher zero-shot AUC/F1 is $.500/.095$ on board and $.662/.666$ on card, so the current method is a target-adapted contact prior requiring target tactile demonstrations.

TABLE II
CONTACT PREDICTION AUC/F1 USING PHASE, PROGRESS, OR VPCE.

Task	Phase-4	Progress-20	VPCE
SWIPE	$.670/.000$	$.853/.597$	$\mathbf{.963/.759}$
WIPE	$.666/.000$	$.743/.406$	$\mathbf{.885/.705}$
INSERT	$.684/.000$	$.808/.539$	$\mathbf{.946/.735}$

Q2: Loss-weight optimization. The concern is correct. The original $w(c) = \mathrm{clip}(1 + \mathrm{softplus}(\gamma)c + \delta, 1)$ is optimized by the same weighted denoising loss, which encourages smaller weights; checkpoint inspection gives $w \approx 1.000$–1.005. The lower bound prevents down-weighting but reduces the term to ordinary DP loss. We therefore replace it with the fixed bounded schedule $w(c) = \mathrm{clip}(1 + c, 1, 2)$. The updated $15/25$ result shows that this correction remains usable, not that loss weighting alone improves over the incompletely aligned Legacy Full control.

Q3: Masking and free-space motion. We agree that Eq. (9) was too coarse: the post-encoder block included proprioception, gripper width, and tactile features. The revision always retains proprioception and gripper width and masks only tactile/contact features; its preliminary result is $13/20$. This preserves closed-loop state feedback, while action-chunk prediction and receding-horizon execution provide additional temporal coherence. We do not claim that the original whole-block mask formally guaranteed smoothness.

Q4: Gating and component ablations. The submitted ablation already showed that removing horizon conditioning or masking reduced SWIPE to $2/10$ and $0/10$, whereas removing gating gave $9/10$. The new no-gating ($8/20$) and bounded-gate ($7/20$) retrains likewise provide no evidence that the tested gate improves binary success; traces indicate that it can over-suppress vision. We will therefore demote gating from the core claim and report it as an auxiliary, potentially negative ablation.

Protocol and scope. All new counts in Table I are operator-held and cannot exclude compliance from the human-held object. We will not pool them with the rigid-mount study (DP-VT, Legacy Full, and fixed-loss VPCE-DP, randomized N=20 each), whose results remain **TBD**. Task-specific thresholds, target adaptation, and the base-frame z-force proxy are limitations; force norms or local-frame multi-axis changes are natural extensions, not current evidence. Finally, SWIPE is the strongest result; with N=20 and overlapping intervals, WIPE/INSERT are supportive trends rather than conclusive gains.